Ibn Tofail University

Algebra IV (S3)

Problem Set II

Exercise 1:

Triangularize the following matrices in $M_3(\mathbb{R})$ when possible.

$$A = \begin{pmatrix} 2 & 0 & 4 \\ 3 & -4 & 12 \\ 1 & -2 & 5 \end{pmatrix} \quad B = \begin{pmatrix} -2 & 2 & -1 \\ -1 & 1 & -1 \\ -1 & 2 & -2 \end{pmatrix} \quad C = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 1 & 0 \\ -1 & 1 & 3 \end{pmatrix}$$

Correction

Exercise 2:

Determine the minimal polynomial of the following matrices:

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \quad C = \begin{pmatrix} 1 & 2 & -2 \\ 2 & 1 & -2 \\ 2 & 2 & -3 \end{pmatrix} \quad D = \begin{pmatrix} 3 & 0 & 8 \\ 3 & -1 & 6 \\ -2 & 0 & -5 \end{pmatrix}$$

Correction

Exercise 3:

Let A be the matrix:

$$A = \begin{pmatrix} 1 & 1-b & b & 0 \\ -a & a & -a & 1+b \\ -a & a-1 & 1-a & 0 \\ 1 & -b & 1 & a \end{pmatrix}$$

where $a, b \in \mathbb{R}$.

1. Give a necessary and sufficient condition for A to be diagonalizable.
2. Using the spectrum of A (the set of eigenvalues of A), show that $A + I_4$ is invertible.
3. Calculate $(A + I_4)^{-1}$ when A is diagonalizable.

Correction

Exercise 4:

Let $A \in M_n(\mathbb{R})$.

1. If A is invertible, show that A is diagonalizable if and only if A^2 is diagonalizable.
2. Does the result still hold if A is not invertible?

Correction

Exercise 5:

Let E be a finite-dimensional $\mathbb{R}$-vector space and f an endomorphism of E such that:

$$(f - id_E)^3 \circ (f + 2id_E) = 0$$

$$(f - id_E)^2 \circ (f + 2id_E) \neq 0$$

Is f diagonalizable?

Correction

Exercise 6:

Let $n \in \mathbb{N}^*$ and $A \in M_n(\mathbb{R})$ such that $A^4 = 7A^3 - 12A^2$. Show that A is triangularizable in $M_n(\mathbb{R})$, that $\mathrm{tr}(A) \in \mathbb{N}$ and that $\mathrm{tr}(A) \leq 4n$.

Correction

Exercise 7:

Let $n \in \mathbb{N} \setminus \{0, 1\}$ and $A = (a_{ij}) \in M_n(\mathbb{R})$ with $a_{1j} = a_{j1} = 1$ for $j > 1$ and $a_{11} = a_{ij} = 0$ for $i, j > 1$.

1. Calculate A^3 and deduce a simple relation between A^3 and A.
2. Deduce that A is diagonalizable and give its eigenvalues.
3. Diagonalize A.

Correction

Exercise 8:

Let $A \in M_3(\mathbb{R})$ such that $A^3 + A = 0$ and $A \neq 0$.

1. Determine $\text{spect}(A)$ (the spectrum of A).
2. Is A diagonalizable in $M_3(\mathbb{R})$?
3. Is A diagonalizable in $M_3(\mathbb{C})$?
4. Show that A is similar to: $\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}$ in $M_3(\mathbb{R})$.

Correction

Exercise 9:

Let $A = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}$ where $n \in \mathbb{N}^*$.

1. Calculate A^2.
2. Is A diagonalizable:
   (a) In $M_{2n}(\mathbb{R})$?
   (b) In $M_{2n}(\mathbb{C})$?

   If yes, perform the diagonalization.

Correction

Exercise 10:

Let E be a finite-dimensional $\mathbb{R}$-vector space and f an endomorphism of E such that $f^2 = f$ and $f \neq 0$. How should t be chosen so that $(id_E + tf)$ is invertible? Calculate $(id_E + tf)^{-1}$.

Correction